

Lecture Notes: Introduction to Voting Theory

Math F113X

Majority vs Plurality, Preference Schedule

Definition: **Plurality Winner**

The person who gets the most votes is the winner!
Regardless of their total vote percentage

If there are 17 candidates and votes are distributed equally,
 $\frac{1}{17} \approx 5.9\%$

Definition: **Majority Winner**

A candidate who receives more than 50% of the vote
(50% + 1 vote)

Who wins in an election?

Ten Alaskans are asked to vote on the “best” of four Alaskan villages.

Voters: Bishop, Claman, Dunbar, Giessel, Hughes, Kawasaki,
Myers, Olson, Tobin, Wilson

Villages: **A**dak, **B**ettles, **C**hevak, **D**iomedede

Given the vote below, who wins? Did they win a **majority**?

plurality winner!

Vote %

$$A = \frac{4}{10} = 0.4$$
$$= 40\%$$

$$B = 30\%$$
$$C = 20\%$$
$$D = 10\%$$

voter	Bishop	Claman	Dunbar	Giessel	Hughes	Kawasaki	Myers	Olson	Tobin	Wilson
vote	A	A	A	A	B	B	B	C	C	D

A : 4 votes C : 2 votes

B : 3 votes D : 1 vote

$$\text{Total vote \%} = \frac{\# \text{ votes}}{\# \text{ possible}} (100)$$

Who wins in an election?

Ten Alaskans are asked to vote on the “best” of four Alaskan villages.

Voters: Bishop, Claman, Dunbar, Giessel, Hughes, Kawasaki, Myers, Olson, Tobin, Wilson

Villages: Adak (A), Bettles (B), Chevak (C), Diomedede (D)

Suppose, in a different world, they voters voted this way. Now what?

voter	Bishop	Claman	Dunbar	Giessel	Hughes	Kawasaki	Myers	Olson	Tobin	Wilson
vote	A	A	A	B	B	B	C	C	C	D

$$\begin{array}{ll} A = 3 = 30\% & C = 3 = 30\% \\ B = 3 = 30\% & D = 1 = 10\% \end{array} \quad \left. \vphantom{\begin{array}{ll} A = 3 = 30\% \\ B = 3 = 30\% \end{array}} \right\} \rightarrow \text{tie!}$$

Who wins in an election?

Ten Alaskans are asked to vote on the “best” of four Alaskan villages.

One option is to **collect more information**. See the new vote tally.

voter	Bishop	Claman	Dunbar	Giessel	Hughes	Kawasaki	Myers	Olson	Tobin	Wilson
1st choice	A	A	A	B	B	B	C	C	C	D
2nd choice	C	C	C	C	C	C	D	D	B	C
3rd choice	B	B	B	D	D	D	B	B	A	B
4th choice	D	D	D	A	A	A	A	A	D	A

Preference Schedule

# votes	3	3	2	1	1
1st choice	A	B	C	C	D
2nd choice	C	C	D	B	C
3rd choice	B	D	B	A	B
4th choice	D	A	A	D	A

“Fair” voting systems

Fairness Criteria

Statements that seem like they should be true in a fair election

Condorcet Criterion

The winner of an election should win every head-to-head matchup that they participate in!

Condorcet Winner

Candidate who wins the most head-to-head matchups

Condorcet Winner

Show that Chevak (C) is the Condorcet Winner in the vote tally below

# votes	3	3	2	1	1
1st choice	A	B	C	C	D
2nd choice	C	C	D	B	C
3rd choice	B	D	B	A	B
4th choice	D	A	A	D	A

A vs B: 3 vs 7, **B wins**

A vs C: 3 vs 7, **C wins**

A vs D: 4 vs 6, **D wins**

B vs C: 3 vs 7, **C wins**

B vs D: 7 vs 3, **B wins**

C vs D: 9 vs 1, **C wins**

A vs B

# votes	3	3	2	1	1
1st choice	A	B	C	C	D
2nd choice	C	C	D	B	C
3rd choice	B	D	B	A	B
4th choice	D	A	A	D	A

A vs C

# votes	3	3	2	1	1
1st choice	A	B	C	C	D
2nd choice	C	C	D	B	C
3rd choice	B	D	B	A	B
4th choice	D	A	A	D	A

A vs D

# votes	3	3	2	1	1
1st choice	A	B	C	C	D
2nd choice	C	C	D	B	C
3rd choice	B	D	B	A	B
4th choice	D	A	A	D	A

B vs C

# votes	3	3	2	1	1
1st choice	A	B	C	C	D
2nd choice	C	C	D	B	C
3rd choice	B	D	B	A	B
4th choice	D	A	A	D	A

B vs D

# votes	3	3	2	1	1
1st choice	A	B	C	C	D
2nd choice	C	C	D	B	C
3rd choice	B	D	B	A	B
4th choice	D	A	A	D	A

C vs D

# votes	3	3	2	1	1
1st choice	A	B	C	C	D
2nd choice	C	C	D	B	C
3rd choice	B	D	B	A	B
4th choice	D	A	A	D	A

C won all it's head-to-head matchups! So C is the **Condorcet Winner**